

Technology Stack

Date	29 June 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well Being
Maximum Marks	2 Marks

Define Your Technology Stack:

The HDI Prediction System is developed using Python, Flask, HTML, CSS, JavaScript, and NumPy with a trained machine learning model stored as `model.pkl`.

The technology stack enables efficient data processing, real-time HDI prediction, and an interactive web interface.

The selected tools ensure the application is simple, maintainable, accurate, and scalable for future enhancements.

Technology Stack Details

S. No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Jinja2 Templates	Provides a responsive and user-friendly web interface for entering HDI parameters and displaying prediction results dynamically.

2	Backend / Server-Side	Python, Flask Framework	Flask handles routing, form submission, prediction requests, model loading, and communication between the frontend and the machine learning model.
3	Database / Data Storage	Pickle (model.pkl)	Stores the trained machine learning model efficiently, allowing it to be loaded quickly for real-time predictions without retraining.
4	Cloud / Hosting / Deployment	Flask Development Server (Localhost)	Used to run and test the application locally during development. The application can later be deployed to cloud platforms such as Render, Heroku, or AWS.
5	Version Control & CI/CD	Git, GitHub	Enables version control, project backup, collaboration, and easy management of code changes throughout development.
6	Third-Party APIs / Other Tools	NumPy, Pickle	NumPy processes input data into arrays required by the machine learning model, while Pickle serializes and loads the trained model for prediction.